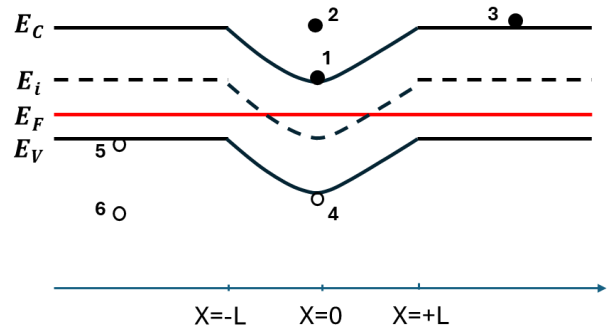


Quiz 2 (20mins)

Q1. For two hydrogen atoms, plot the potential energy and mention the status of allowed energy states with potential energy when both the atoms are very close such that all the energy states are shared with both the atoms. What is the condition for stable co-valent bond formations? Describe using kinetic energy and potential energy with the importance of lattice constant. (3)

Q2 Consider the following energy band diagram. Take the semiconductor represented to be Si maintained at 300 K with $E_i - E_F = E_g/4$ at $x = \pm L$ and $E_F - E_i = E_g/4$ at $x = 0$. Note the choice of E_F as the energy reference level. Find the K.E. and P.E. of the electrons and holes pictured on the diagram. (3)



Note: No partial marks. Every charge carrier should have correct KE and PE.

Q3 a) What is an intrinsic semiconductor? (1)

b) At 0K, are there any free charge carriers? (1)

c) Describe any two methods to increase the charge carriers in the semiconductor? Which method is feasible to increase the free charge carriers? (2)